

CO4-AT2-CODING GOLF CHALLENGE

Name: V. Madhumitha

Reg no: 192521332

Course name: Data structure
CSA0311

Course code:

Q1: Bubble sort

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Madhu Miitha V
192521332RB

QUESTIONS

Q1
Bubble Sort
50 marks

Q2
Insertion Sort
50 marks

2/2 answered

Q1 Bubble Sort 50 marks

Write a program that implements the bubble sort

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n, a[100], i, j, t, swapped;
4     scanf("%d", &n);
5
6     for (i = 0; i < n; i++)
7         scanf("%d", &a[i]);
8
9     for (i = 0; i < n - 1; i++) {
10        swapped = 0;
11        for (j = 0; j < n - i - 1; j++)
12            if (a[j] > a[j+1]) {
13                t = a[j];
14                a[j] = a[j+1];
15                a[j+1] = t;
16                swapped = 1;
17            }
18        if (!swapped)
19            break;
20    }
21
22    for (i = 0; i < n; i++) printf("%d ", a[i]);
23    return 0;
24 }
```

Submitted

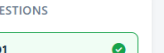
Input

5
64 32 25 12 22

Output

12 22 25 32 64

Q2. Insertion sort

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QUESTIONS

Q1
Bubble Sort
50 marks

Q2
Insertion Sort
50 marks

2/2 answered

Q2 Insertion Sort 50 marks

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[100], n, i, j, key;
4     scanf("%d", &n);
5     for (i = 0; i < n; i++)
6         scanf("%d", &a[i]);
7
8     for (i = 1; i < n; i++) {
9         key = a[i];
10        j = i - 1;
11        while (j >= 0 && a[j] > key) {
12            a[j + 1] = a[j];
13            j--;
14        }
15        a[j + 1] = key;
16    }
17
18    for (i = 0; i < n; i++)
19        printf("%d ", a[i]);
20    return 0;
21 }
```

Submitted

Input

5
9 5 1 4 3

Output

1 3 4 5 9